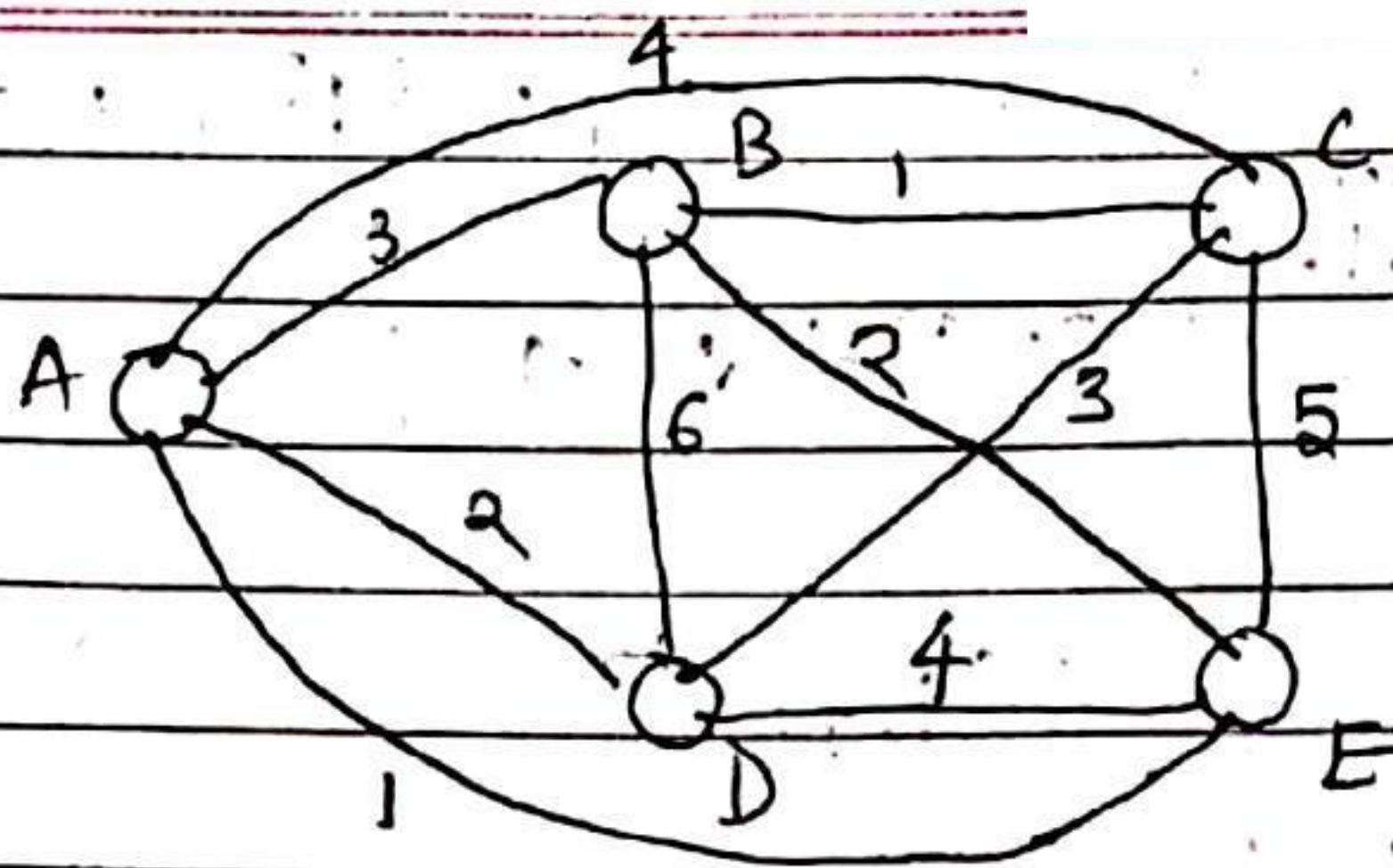


Travelling Salesman problem

atlantis

Date _____

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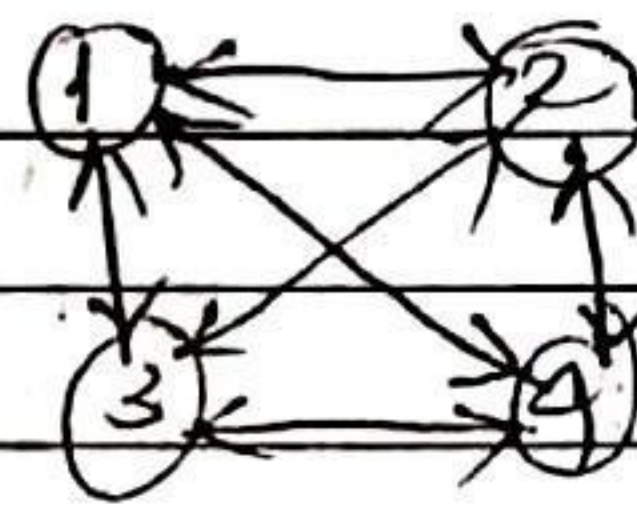
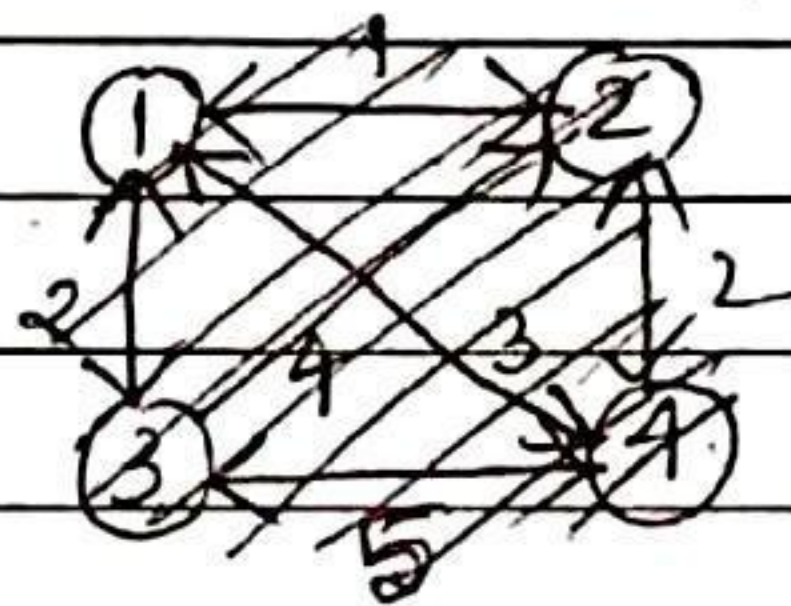
$(A)BCDEA$
 $ACBDEA$
 $ADBCEA$

$(n-1)!$

$O(n!) \gg 2^n$

• Detecting Optimal Substructure -

Directed graph -



Cost matrix =

	1	2	3	4
1	0	1	2	3
2	1	0	4	2
3	1	2	0	5
4	3	4	1	0

Time-T

$T(1, \{2, 3, 4\}) = \min$

5

$\checkmark (1, 2) + T(2, \{3, 4\})$
 $(1, 3) + T(3, \{2, 4\})$
 $(1, 4) + T(4, \{2, 3\})$

$T(2, \{3, 4\}) = \min$

4

$(2, 3) + T(3, \{4\})$
 $\checkmark (2, 4) + T(4, \{3\}) \checkmark = 1$

$\checkmark (4)$

$$T(3, \{2, 4\}) = \min \begin{cases} (3, 2) + T(2, \{4\}) = 7 \\ (3, 4) + T(4, \{2\}) = 5 \end{cases}$$

$$T(4, \{2, 3\}) = \min \begin{cases} (4, 2) + T(2, \{3\}) = 5 \\ (4, 3) + T(3, \{2\}) = 4 \end{cases}$$

$$T(3, \{4\}) = (3, 4) + T(4, \emptyset) = 8$$

$$T(4, \{3\}) = (4, 3) + T(3, \emptyset) = 2$$

$$T(2, \{4\}) = (2, 4) + T(4, \emptyset) = 5$$

$$T(4, \{2\}) = (4, 2) + T(2, \emptyset) = 5$$

$$T(2, \{3\}) = (2, 3) + T(3, \emptyset) = 5$$

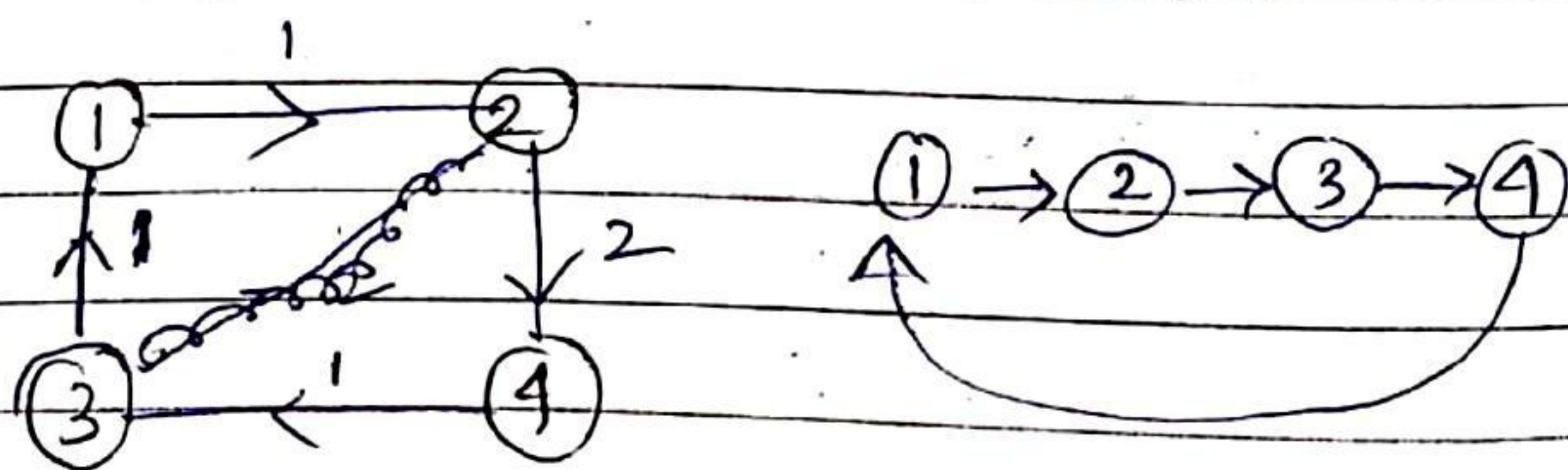
$$T(3, \{2\}) = (3, 2) + T(2, \emptyset) = 3$$

$$T(2, \emptyset) = (2, 1)$$

$$T(3, \emptyset) = (3, 1)$$

$$T(4, \emptyset) = (4, 1)$$

minimum is 5; To take from ① to {2, 3, 4} and
 Comeback ① - minimum cost is 5.



• Bottom up dynamic programming algorithm -

Recursive equation -

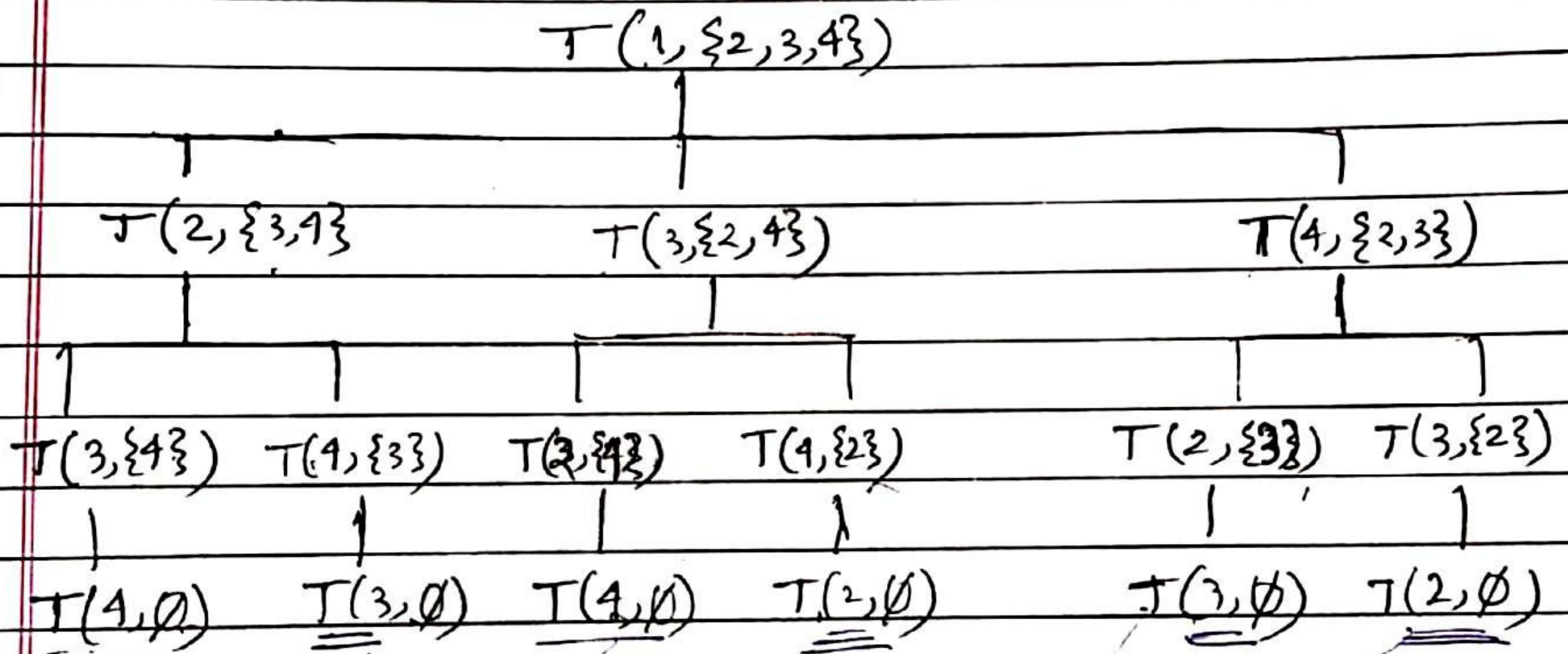
$S \rightarrow$ set of nodes.
 $i \rightarrow$ starting point.

$$T(i, S) = \min_{j \in S} ((i, j) + T(j, S - \{i\})) ; S \neq \emptyset$$

$$= (i, i) ; S = \emptyset$$

Recursive tree -

$$V = (1, 2, 3, 4)$$



Total sub problem = 15

total unique subproblem are = 13

Formula = $(n-1) 2^{n-2}$

$$= (4-1) 2^2$$

$$= 3 \times 4 = 12 \rightarrow \text{if apply Dynamic-P}$$

12 is the table size.
will be the

Time complexity = $O(n 2^n)$ $\neq O(n)$

$$= O(n^2 2^n)$$

Space complexity = $O(n^2 2^n)$